CHANGELOG - version 2 (after the instructor's checklist)

Version 1 PDFs were printed before the instructor's checklist arrived. This list says exactly what changed so you can reprint only what you need. Page numbers refer to the NEW PDFs in `PDF_Print/`.

Why it changed

The instructor's checklist for the practical assessment names: pandas **merge**, **duplicates**, **shape**, **save to CSV**; matplotlib **box plots**; SQLite `to_sql`, **UPDATE**, **export results to CSV**; PyTorch **seed**, `y = X @ w + b + noise`, **shapes**, regression with Linear + ReLU. Version 1 covered some of these thinly and some not at all, and advised retyping a whole solution, which contradicts the instructor's "practise the individual operations".

Documents you must REPLACE (reprint fully)

Document	Pages (old -> new)	What changed
`Exam_Day_Cheat_Sheet`	3 -> 4	Rewritten (v2). Added: shape, `duplicated()` and `drop_duplicates(subset=)`, **merge** (inner/left/right/outer, `left_on/right_on`, `indicator`), filter/select/sort, `to_csv`; **scatter** and **box plot**; `to_sql`, **UPDATE** with `rowcount`, `?` parameters, **export query results to CSV** (2 ways); **seeded regression network** (`y = X @ w + b + noise`, no sigmoid) replacing the old classification block; new error rows (merge dtype, `(N,N)` shape bug, bindings, UPDATE without commit). Whole script tested twice. A bug found during testing was fixed (`cursor.description` is `None` after an UPDATE).

New documents (print if you want them)

Document	Pages	What it is
`Topics/T9_Checklist_Operation_Cards`	17	27 small "operation cards", one per checklist bullet: analogy, minimal code, what each part does, check, common slip, self-test. Plus a tracker and answer key.
`Mock_Exams/MockC/*`	4 + 7	A mock written to the checklist (two CSVs to merge, duplicates, 4 plots, SQLite with UPDATE and export, seeded regression).
`Solutions_py/practical_operation_drills.py`	-	all 27 cards as runnable functions with PASS/FAIL
`Solutions_py/pytorch_regression_practical.py`	-	the full regression program, heavily commented

READY-MADE PRINT FILES for the new pages

The folder `Print_New_Pages/` holds exactly the pages listed below as three small PDFs (42 pages in total), each page stamped top-right with its document and page: - `Print_1_SQLite_T4_new_pages.pdf` (10 pages) - `Print_2_Pandas_and_Matplotlib_T5_T6_new_pages.pdf` (18 pages) - `Print_3_CheatSheet_v2_and_PyTorch_T7_new_pages.pdf` (14 pages, includes the new cheat sheet) Page 1 of each document replaces your old page 1; the other new pages go BEHIND your old copy, in order. `T2` and `T3` did not change, so there is nothing new to print for them. Rebuild with `_tools/make_print_extracts.py`.

Documents you can KEEP and just add pages to

These only had material ADDED (plus two small corrections marked below). Reprint the pages shown and slot them in.

Document	New length	Reprint these pages	What is new
`T4_SQLite`	11 -> 18	**1**, **10-18**	p1: instructor checklist box. p10: 12 new trap rows (UPDATE without WHERE, rowcount, NumPy values in WHERE, `LIKE`, export index column). p11: Q11-Q16. p12: extra quick-reference box and **new section 8**: 8.1 connect, 8.2 `to_sql`, 8.3 SELECT/WHERE/ORDER BY/LIMIT, 8.4 **UPDATE** (full worked), 8.5 `?` parameters, 8.6 commit, 8.7 **export to CSV (3 ways)**, 8.8 ten statements to know.
`T5_Pandas_NumPy`	10 -> 15	**1**, **5-8**, **11-15**	p1: checklist box + **CORRECTION** (below). p5-8: new 3.11 shape, 3.12 duplicates, **3.13 merge**, 3.14 select/filter, 3.15 save to CSV. p11-15: traps 5.1, questions Q10-Q15, extra quick-reference lines.
`T6_Matplotlib`	11 -> 14	**1**, **6-8**, **11-14**	p1: checklist box + **CORRECTION** (below). p6: 3.11 scatter/line/bar templates. p7: **3.12 box plots**. p8: 3.13 which plot for which question. p11-14: traps 5.1, questions Q11-Q17, extra quick-reference lines.
`T7_PyTorch_Basics`	11 -> 17	**1**, **6-11**, **15-17**	p1: instructor checklist (1b). p6-11: **3.8 REGRESSION step by step** (seed + reproducibility proof, data and shapes, model, loss/optimiser/learning-rate table, loop, plot, evaluation + the single-Linear check that recovers `w` and `b`). p15-17: traps 5.11-5.16, questions Q11-Q17, extra quick-reference box.
`00_Exam_Pattern_Analysis`	4 -> 5	**1** (section 0), **3** (section 4), **5** (section 8)	new section 0 (checklist map); likelihood table re-weighted to the checklist; study plan changed to operations-first.
`Paper_Solutions/P1_2024_Paper_Explained`	12 -> 12	**1**	one note at the top: the 2024 paper uses a sigmoid network; your assessment is regression; read P1 for understanding, do not memorise.
`Mock_Exams/README_How_to_use_mocks`	2 -> 3	**2-3**	Mock C section appended.

Corrections of earlier text (old pages contain these errors)

1. **T5 page 1 / T6 page 1:** the 2024 `sales_data.csv` has **6 exact duplicate rows** (04/04/2022 to 10/04/2022 appear twice). Earlier T5/T6 text did not say so, and the "expected numbers" in T6 template K (Biscuits 203, Orange Juice 248, Sweets 261, ...) were computed WITH the duplicates left in. After `drop_duplicates()` the quantities are Biscuits 196, Chocolate 120, Orange Juice 232, Sweets 252, Unknown 33, Water 90 (the 2023 chart is unchanged).
2. **T4,** `(username)` **trap row:** the message "uses 1, and there are 2 supplied" now says N, because the number depends on the length of the string.
3. **Cheat sheet v1:** its PyTorch block (sigmoid output, random targets) is replaced by the regression recipe, and its `w` shape error row now says the shape silently becomes `(N, N)`.

Documents with NO change

`T1`, `T2`, `T3`, `T8`, `P2`, `P3`, `P4`, Mock A and B papers and marking schemes.

One version warning that applies to everything

Naming box-plot boxes: `labels=` was **removed in matplotlib 3.11**; `tick_labels=` only exists in **3.9+**. The version-proof way, used in the cheat sheet, T9 and Mock C, is `plt.boxplot(groups)` then `plt.xticks(range(1, n + 1), names)`. (T6 shows all three and explains the differences.)